Vectors

Ch 3 in HRW

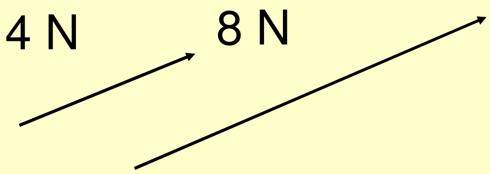
Scalars and Vectors

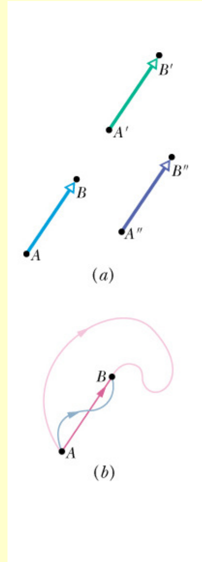
- A *scalar* quantity is one that can be described by a single number and the unit. It has magnitude but no direction.
  - temperature, speed, mass
- A *vector* is a physical quantity which has both with both magnitude and direction:
  - velocity, force, displacement
  - $\bar{F}$ or $\vec{F}$ or $\boldsymbol{F}$
  - Magnitude of the vector: $\left|\vec{F}\right|$

$$\therefore F = \left|\vec{F}\right|$$

Scalars and Vectors

- Arrows are used to represent vectors. The direction of the arrow gives the direction of the vector.
- By convention, the length of a vector arrow is proportional to the magnitude of the vector.





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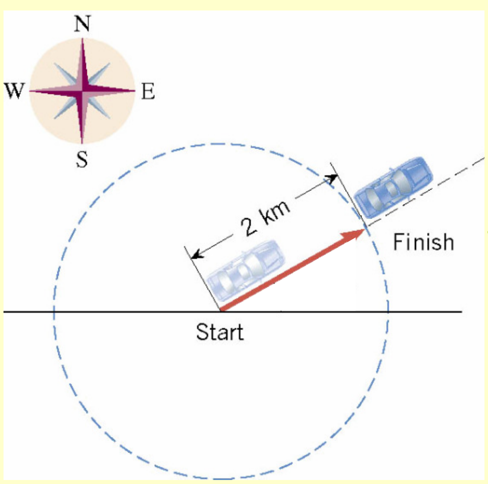
Scalars and Vectors



- The car moved a distance of 2 km.
- The car moved a distance of 2 km in a direction 30° north of east.

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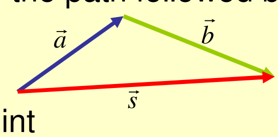
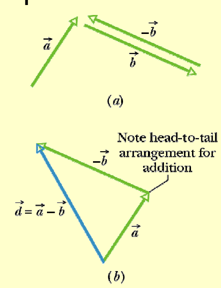
Vector addition-Geometric

- A particle moves from point A to point B, then from point B to point C.
- Overall displacement irrespective of the path followed by the particle.
- Head to tail method
- Resultant = Starting point to End point
- Commutative law
- Associative law
- Vector subtraction





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Checkpoint 1

- The magnitudes of displacements $\vec{a}$ and $\vec{b}$ are 3m and 4 m respectively, and $\vec{c} = \vec{a} + \vec{b}$ Considering the various orientations of $\vec{a}$ and $\vec{b}$, what is the (i) maximum and (ii) minimum possible magnitude for $\vec{c}$?

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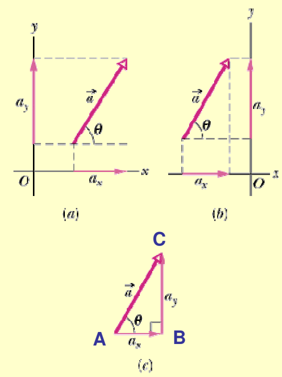
Components of vectors

- A component of a vector is a projection of the vector on an axis.
- a_x is the component of vector $\vec{a}$ along the x-axis, and a_y is the component of vector along the y-axis.
- Draw $\perp$ lines to axes.
- Resolve the vector.
- a_x, a_y both positive in this case.
- Generally, 3 components x-, y- and z-components.



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Components of vectors

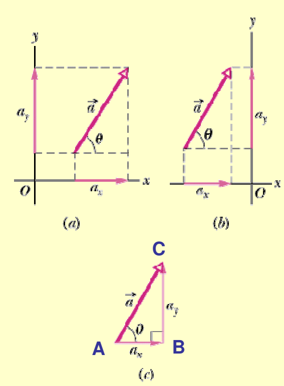
- Usually:

$$\cos\theta = \frac{adj}{hyp} \qquad \text{then } a_x = a\cos\theta$$

$$\sin\theta = \frac{opp}{hyp} \qquad \text{then } a_y = a\sin\theta$$

- $$a = \sqrt{a_x^2 + a_y^2}$$

$$\tan\theta = \frac{a_y}{a_x}$$



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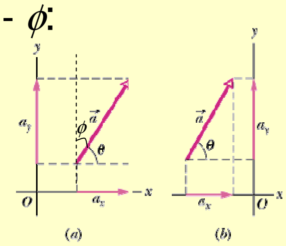
Components of vectors

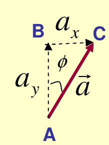
- Now if you use the other angle - ϕ:

$$\cos\phi = \frac{adj}{hyp} \qquad \text{then } a_x = a\cos\phi$$

$$\sin\phi = \frac{opp}{hyp} \qquad \text{then } a_y = a\sin\phi$$

$$a = \sqrt{a_x^2 + a_y^2}$$





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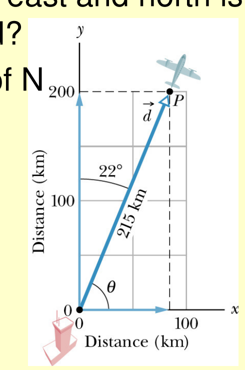
$$\tan\theta = \frac{a_y}{a_x}$$

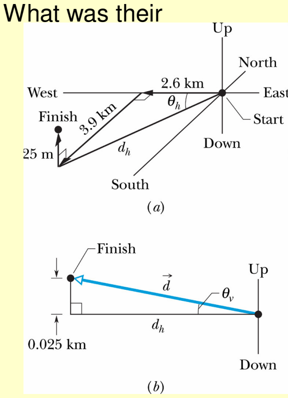
Sample Problem p42

- A small plane leaves an airport on an overcast day and is later sighted 215 km away in a direction making an angle of 22° east of due north. How far east and north is the plane from the airport when sighted?
- Given: distance: 215 km, angle 22° E of N
- Sketch
-



Problem

- For 2 decades spelunking teams sought a connection between Flint Ridge cave system and Mammoth Cave (Kentucky, USA) When the connection was finally discovered, the combined system was declared the world's longest cave (>200km) The team that found the connection had to crawl, climb and squirm through many passages, travelling a net 2.6 km westward, 3.9 km southward and 25m upward. What was their displacement from start to finish?



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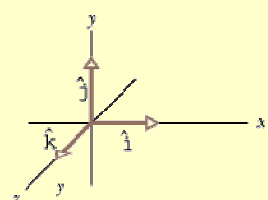
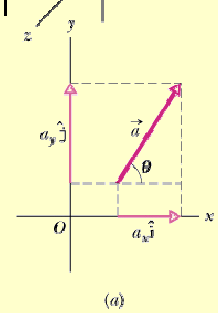
Unit Vectors



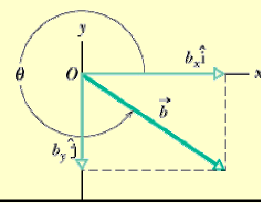
- A unit vector has a magnitude of exactly 1 and points in a particular direction.
- Dimensionless and unitless
- Purpose is to specify direction
- Labelled: $\hat{i}$, $\hat{j}$ and $\hat{k}$ for x, y, z direction
- Right handed co-ordinate system.

$$\vec{a} = a_x\hat{i} + a_y\hat{j}$$

$$\vec{b} = b_x\hat{i} + b_y\hat{j}$$



- Vector components
  - $a_x\hat{i}$ and $a_y\hat{j}$, $b_x\hat{i}$ and $b_y\hat{j}$
- Scalar components
  - a_x and a_y, b_x and b_y



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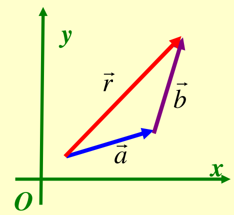
Adding Vectors by Components

- Consider the vectors

$$\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$
$$\vec{b} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$$
$$\vec{r} = \vec{a} + \vec{b}$$



- Then

$$r_x = a_x + b_x$$
$$r_y = a_y + b_y$$
$$r_z = a_z + b_z$$

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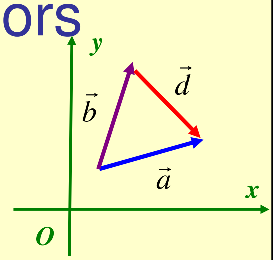
Subtraction of Vectors



- Consider the vectors

$$\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$$
$$\vec{b} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$$
$$\vec{d} = \vec{a} - \vec{b}$$

- Then

$$d_x = a_x - b_x$$
$$d_y = a_y - b_y$$
$$d_z = a_z - b_z$$

- and

$$\vec{d} = d_x\hat{i} + d_y\hat{j} + d_z\hat{k}$$

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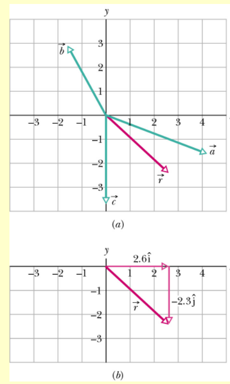
Sample Problem p45

- Consider the three vectors as shown in the figure.

$$\vec{a} = (4.2 \text{ m})\hat{i} - (1.5 \text{ m})\hat{j}$$
$$\vec{b} = (-1.6 \text{ m})\hat{i} + (2.9 \text{ m})\hat{j}$$
$$\vec{c} = (-3.7 \text{ m})\hat{j}$$

What is their vector sum (also shown)?



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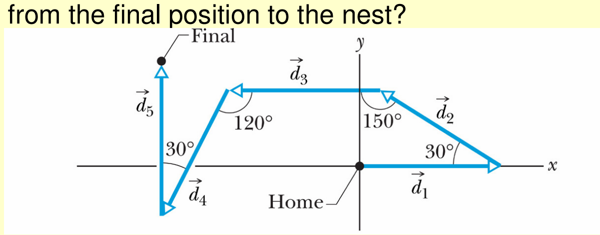
Sample Problem p45-46

- When an ant wants to return to its home next, it effectively sums its displacements along the axes of the system to calculate a vector that points directly home. Consider an ant making five runs of 6.0 cm each as shown in figure.
  - (i) What are the magnitude and direction of the ants net displacement vector?
  - (ii) what are the magnitude and direction of the homeward vector from the final position to the nest?



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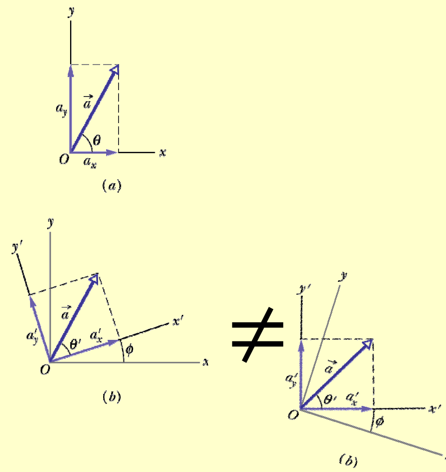
Vectors and Physics

- Usually have shown axes for vectors parallel to the page edges.
- No particular reason for this.
- Can rotate the axis and still have the “same” resultant vector
- Use this a lot in Physics particularly



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Problems

- 8. A Person walks in the following pattern: 3.1 km north, then 2.4 km west, and then 5.2 km south. (a) Sketch the vector diagram that represents this motion. (b) How far is he from his starting point? (c) In what direction is he w.r.t. his starting point?
- 25. Oasis *B* is 25 km east of Oasis *A*. Starting from oasis *A* a camel walks 24 km in a direction 15° S of E and then 8 km N. How far is the camel from Oasis *B*?
- 26. What is the sum of the following 4 vectors in unit-vector notation?

$\vec{A} = (2.00\text{ m})\hat{i} + (3.00\text{ m})\hat{j}$ $\qquad$ $\vec{B}: 4.00$ m, at $+65.0°$

$\vec{C} = (-4.00\text{ m})\hat{i} + (-6.00\text{ m})\hat{j}$ $\qquad$ $\vec{D}: 5.00$ m, at $-235°$

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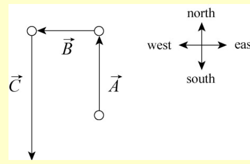
Solutions

- 8.
- (b) $|\vec{r}| \approx 3.2 \text{ km}$



- (c) There are two possibilities for the angle:

$$\theta = \tan^{-1}\left(\frac{-2.1 \text{ km}}{-2.4 \text{ km}}\right) = 41.2° \text{ or } 221.2°$$

Which one do we choose???

- 26. (a) $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$

$$\vec{R} = (-3.18 \text{ m})\hat{\text{i}} + (4.72 \text{ m})\hat{\text{j}}$$

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Multiplying Vectors

- Multiplying a vector by a scalar
  - new vector s (magnitude of vector)
  - Direction is same as original vector if s is positive
  - Divide vector by s = multiply by $1/s$.
- Multiplying a vector by a vector
  - Two methods
  - One method produces a scalar – scalar product
  - One method produces another vector – vector product

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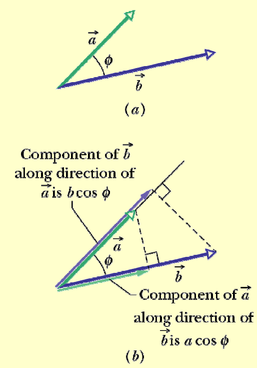
Scalar product

- Two vectors $\vec{a}$ and $\vec{b}$
- Scalar product or dot product

  $$\vec{a} \cdot \vec{b} = ab\cos\phi$$

  - a, b magnitudes of the vectors
  - ϕ – angle between the directions of two vectors
- The product of the magnitude of one vector and the scalar component of the other vector along the direction of the first. (see (b) in figure)
- If ϕ = 0° then dot product is a max.
- If ϕ = 90° then dot product is 0.
- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$



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Scalar Product

- Scalar product using unit vector notation

$$\begin{aligned}
\vec{a} \cdot \vec{b} &= \left(a_x\hat{i} + a_y\hat{j} + a_z\hat{k}\right) \cdot \left(b_x\hat{i} + b_y\hat{j} + b_z\hat{k}\right) \\
&= \left(a_x\hat{i} \cdot b_x\hat{i}\right) + \left(a_x\hat{i} \cdot b_y\hat{j}\right) + \left(a_x\hat{i} \cdot b_z\hat{k}\right) \\
&\quad + \left(a_y\hat{j} \cdot b_x\hat{i}\right) + \left(a_y\hat{j} \cdot b_y\hat{j}\right) + \left(a_y\hat{j} \cdot b_z\hat{k}\right) \\
&\quad + \left(a_z\hat{k} \cdot b_x\hat{i}\right) + \left(a_z\hat{k} \cdot b_y\hat{j}\right) + \left(a_z\hat{k} \cdot b_z\hat{k}\right) \\
&= a_x b_x + a_y b_y + a_z b_z
\end{aligned}$$

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Checkpoint 4

- Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of the vectors if the dot product equals (i) 0, (ii) 12 units, and (iii) -12 units?

Sample Problem p49

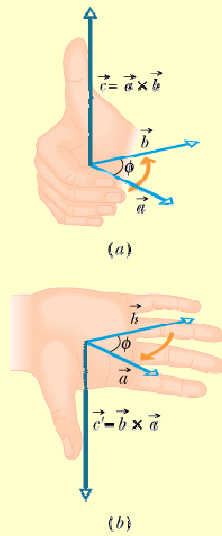
- What is the angle ϕ between $\vec{a} = 3.0\hat{i} - 4.0\hat{j}$ and $\vec{b} = -2.0\hat{i} + 3.0\hat{k}$?

Vector Product

- Two vectors $\vec{a}$ and $\vec{b}$
- Vector product or cross product

  $$c = \left|\vec{a}\times\vec{b}\right| = ab\sin\phi$$

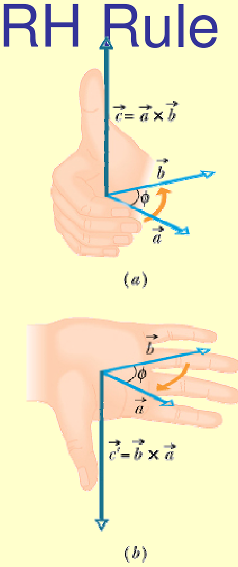
  - c - Magnitude of the third vector
  - a, b magnitudes of two initial vectors
  - ϕ – smaller of the two angle between the two vectors.
- If ϕ = 0° then cross product is 0.
- If ϕ = 90° then cross product is a max.
- Direction of $\vec{c}$ is $\perp$ to plane containing $\vec{a}$ and $\vec{b}$
- RIGHT-hand rule:



Vector Product – RH Rule

- $\vec{c} = \vec{a}\times\vec{b}$
- Place the vectors tail to tail.
- Point the fingers of your right hand in the direction of the first vector.
- your fingers, through the smallest angle between $\vec{a}$ and $\vec{b}$, from vector $\vec{a}$ to $\vec{b}$.
- Your outstretched thumb points in the direction of $\vec{c}$.
- Commutative law does NOT apply.

  $$\vec{a}\times\vec{b} = -\left(\vec{b}\times\vec{a}\right)$$



Vector Product

- $\vec{c} = \vec{a} \times \vec{b}$
- $\vec{a} \times \vec{b} = \left(a_x\hat{i} + a_y\hat{j} + a_z\hat{k}\right) \times \left(b_x\hat{i} + b_y\hat{j} + b_z\hat{k}\right)$

$$= \left(a_x\hat{i} \times b_x\hat{i}\right) + \left(a_x\hat{i} \times b_y\hat{j}\right) + \left(a_x\hat{i} \times b_z\hat{k}\right)$$

$$+ \left(a_y\hat{j} \times b_x\hat{i}\right) + \left(a_y\hat{j} \times b_y\hat{j}\right) + \left(a_y\hat{j} \times b_z\hat{k}\right)$$

$$+ \left(a_z\hat{k} \times b_x\hat{i}\right) + \left(a_z\hat{k} \times b_y\hat{j}\right) + \left(a_z\hat{k} \times b_z\hat{k}\right)$$

$$= \left(a_yb_z - b_ya_z\right)\hat{i} + \left(a_zb_x - b_xa_z\right)\hat{j} + \left(a_xb_y - b_ya_x\right)\hat{k}$$

- Determinant method

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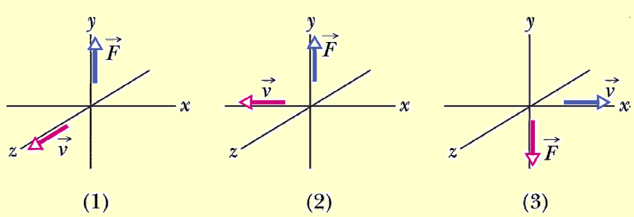
Question 9

- If $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$ and $\vec{v}$ is perpendicular to $\vec{B}$, then what is the direction of $\vec{B}$ in each of the three situations below:



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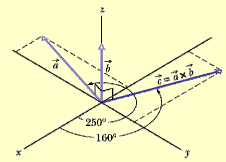
Sample Problem p51

- The figure shows vector $\vec{a}$ lying in the xy plane, with a magnitude of 18 units and pointing in the direction of 250° from the $+x$-direction. Also, vector $\vec{b}$ has a magnitude of 12 units and points along the $+z$ direction. What is the vector product $\vec{c} = \vec{a} \times \vec{b}$?



- Magnitude:
  - Direction:

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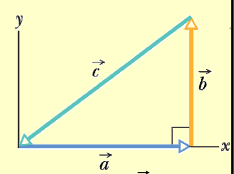
Problems



- 33. For the vectors in the figure, with a = 4, b = 3 and c = 5, what are (a) magnitude and direction of $\vec{a} \times \vec{b}$ (b) magnitude and direction of $\vec{a} \times \vec{c}$ and (c) magnitude and direction of $\vec{c} \times \vec{b}$
- 38. For the following 3 vectors, what is $3\vec{C} \cdot \left(2\vec{A} \times \vec{B}\right)$?

$$\vec{A} = 2.00\hat{i} + 3.00\hat{j} - 4.00\hat{k}$$

$$\vec{B} = -3.00\hat{i} + 4.00\hat{j} + 2.00\hat{k} \qquad \vec{C} = 7.00\hat{i} - 8.00\hat{j}$$

- 57. If $\vec{B}$ is added to $\vec{A}$, the result is $6.0\hat{i} + 1.0\hat{j}$. If $\vec{B}$ is subtracted from $\vec{A}$, the result is $-4.0\hat{i} + 7.0\hat{j}$. What is the magnitude of $\vec{A}$?

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Solutions

- 33. (a) $\left|\vec{a}\times\vec{b}\right| = 12$
  RHRule: along +*z*-axis (out of page)
  (b) $\left|\vec{a}\times\vec{c}\right| = 12$
  $\theta = \tan^{-1}\dfrac{b}{a} = \tan^{-1}\dfrac{3}{4} = 37°$ $\qquad \phi = 90° + 37° = 127°$
  RHRule: along –*z*-axis (into page)
  (c) $\left|\vec{c}\times\vec{b}\right| = 12$
  RHRule: along -*z*-axis (into page)
- 38. $2\vec{A} \times \vec{B} = 44.0\hat{\mathrm{i}} + 16.0\hat{\mathrm{j}} + 34.0\hat{\mathrm{k}}$
  $3\vec{C}\cdot\left(2\vec{A}\times\vec{B}\right) = 540.$
- 57. $A = \sqrt{(1.0)^2 + (4.0)^2} = 4.1$